Data Visualization & Analytics Lab

Experiment No. : 5

Create basic visualizations using D3.js

Experiment No.5

1. **Aim:** Set up D3.js, load a dataset, and explore selection binding by creating basic visualizations (bar charts, line charts) using D3's data loading functions.

2. **Objectives:**

   - Understand why visualization is an important part of data analysis.
   - Understand the components involved in visualization design.

1. **Course Outcomes:** Students will be able to explore D3.js for interactive data visualizations in web browser.

4. **Hardware / Software Required: HTML, CSS, SVG, D3.js**

5. **Theory:**

Data-Driven Documents (D3/D3.js) is a resource JavaScript library for managing documents based on data. D3 is one of the most effective frameworks to work on data visualization. It allows the developer to create dynamic, interactive data visualizations in the browser with the help of HTML, CSS, and SVG.

Set up the D3.js development environment:

We need the following components −

- D3.js library
- Editor
- Web browser
- Web server

We need to include the D3.js library into your HTML webpage in order to use D3.js to create data visualization.
We can do it in the following two ways −

- Download and Include the D3.js library from your project's folder.
- Include D3.js library from CDN (Content Delivery Network).

```
<script src="https://d3js.org/d3.v4.min.js"></script>
```

D3.js - Selections

It allows us to select one or more elements in a webpage. In addition, it allows us to modify, append, or remove elements in a relation to the pre-defined dataset.

- D3.js helps to select elements from the HTML page using following two methods −
  - **d3.select()** − Selects only one DOM element by matching the given CSS selector. If there are more than one elements for the given CSS selector, it selects the first one only.
  - **d3.selectAll()** − Selects all DOM elements by matching the given CSS selector. If you are familiar with selecting elements with jQuery, D3.js selectors are almost the same.
- The D3.js selection provides the **append()** and the **text()** methods to append new elements into the existing HTML documents.
- D3.js provides various methods, **html(), attr()** and **style()** to modify the content and style of the selected elements.

Data Loading in D3:

- D3 provides the following methods to load different types of data from external files.
  - d3.csv()
  - d3.json()
  - d3.tsv()
  - d3.xml()
- **Data Loading d3.csv()**

  Syntax: d3.csv(url[, row, callback]);
  - The first parameter is the url of .csv file, or webapi.
  - The second optional parameter is a conversion function which allows us to change the representation.
  - The third optional parameter is a callback function which will be executed once the .csv file/data is loaded. It passes parsed data object as a parameter to callback function.

Bar Chart Visualization using D3:

Let's plot the share value of a dummy company, XYZ Foods, over a period from 2011 to 2016.

XYZ.csv

```csv
year,value
2011,45
2012,47
2013,52
2014,70
2015,75
2016,78
```

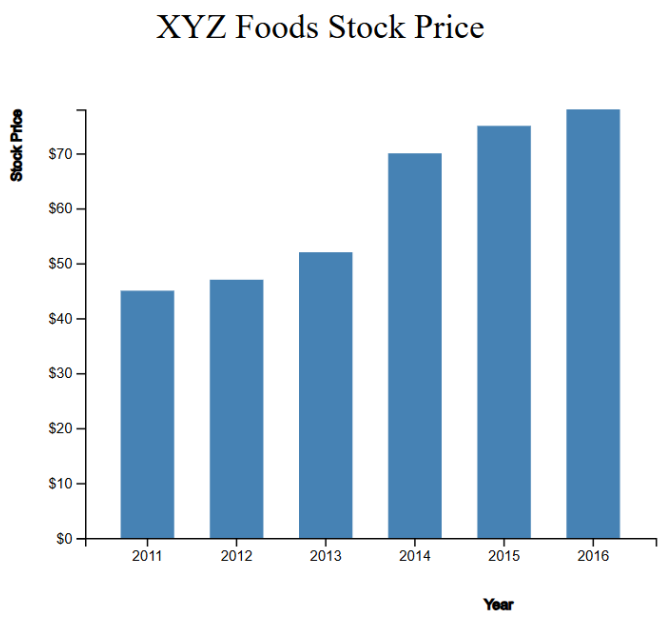
BarchartD3.html

```html
<!doctype html>
<html>
<head>
    <style>
        .bar {
            fill: steelblue;
        }
    </style>
    <script src="https://d3js.org/d3.v4.min.js"></script>
<body>
<svg width="600" height="500"></svg>
<script>

    var svg = d3.select("svg"),
        margin = 200,
        width = svg.attr("width") - margin,
        height = svg.attr("height") - margin

    svg.append("text")
       .attr("transform", "translate(100,0)")
       .attr("x", 50)
       .attr("y", 50)
       .attr("font-size", "24px")
       .text("XYZ Foods Stock Price")

    var xScale = d3.scaleBand().range([0, width]).padding(0.4),
        yScale = d3.scaleLinear().range([height, 0]);

    var g = svg.append("g")
```

```
            .attr("transform", "translate(" + 100 + "," + 100 + ")");

d3.csv("XYZ.csv", function(error, data) {
    if (error) {
        throw error;
    }

    xScale.domain(data.map(function(d) { return d.year; }));
    yScale.domain([0, d3.max(data, function(d) { return d.value; })]);

    g.append("g")
     .attr("transform", "translate(0," + height + ")")
     .call(d3.axisBottom(xScale))
     .append("text")
     .attr("y", height - 250)
     .attr("x", width - 100)
     .attr("text-anchor", "end")
     .attr("stroke", "black")
     .text("Year");

    g.append("g")
     .call(d3.axisLeft(yScale).tickFormat(function(d){
         return "$" + d;
     })
     .ticks(10))
     .append("text")
     .attr("transform", "rotate(-90)")
     .attr("y", 6)
     .attr("dy", "-5.1em")
     .attr("text-anchor", "end")
     .attr("stroke", "black")
     .text("Stock Price");

    g.selectAll(".bar")
     .data(data)
     .enter().append("rect")
     .attr("class", "bar")
     .attr("x", function(d) { return xScale(d.year); })
     .attr("y", function(d) { return yScale(d.value); })
     .attr("width", xScale.bandwidth())
     .attr("height", function(d) { return height - yScale(d.value); });
```

```
    });
</script>
</body>
</html>
```

Output:



5. Conclusion: d3.js is a library used to create visualizations in web browser. We have learnt to selections, data loading and data visualization using html, svg and d3.js

6 Viva Questions:

- What is the use of d3.js?
- What are the functions available in D3 for loading the data?
- What is SVG?

References:

1. The JavaScript library for bespoke data : https://d3js.org/
2. https://www.tutorialspoint.com/d3js/d3js_introduction.htm
3. Video Link for Line Chart Creation in D3: https://www.youtube.com/watch?v=g5bp02-CRAc